

Course Title	Linear Algebra	Course Code	MT1004
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Mathematics & Computing (WK2)	Credit Hrs.	3
Knowledge Area	Mathematics (KA10)	Grading Scheme	Relative
HEC Knowledge Area	Natural Sciences	Applicable From	Spring 2023
Pre-requisite(s)			

Course Objective	Students are expected to learn about system of linear equations, vector spaces, inner products, eigen values and linear transformations. This course also emphasizes the application of linear algebra in science and real life
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No.	Assigned Program Learning Outcome (PLO)
2	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

I = Introduction, R = Reinforcement,

E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Express real-life problems into system of linear equations and solve the system.	A1, Q1, M1	C2	2
2	Recognize vector spaces and/or subspaces and develop their bases.	A2, M2, Q2, F	C2	2
3	Identify inner product spaces and construct orthogonal/orthonormal basis.	Q3, F	C2	2
4	Express a linear transformation graphically/ as matrices and use their properties to solve engineering problems.	A3, F	C2	2

Text Books	Title	Elementary Linear Algebra
	Author	Howard A. Anton (Latest Edition)
	Publisher	
Reference Books	Title	Linear Algebra with Applications
	Author	Bernard Kolman (Latest Edition)
	Publisher	

Week	Course Contents/Topics	Chapter*	CLO*
1	System of Linear Equations and Matrices ,1.1 Introduction to linear system ,1.2 Gaussian Elimination ,1.3 Matrices and Matrix operations	1	1
2	1.4 Inverses; Algebraic properties of Matrices ,1.5 Elementary Matrices and Method of finding matrix inverse	1	1
3	1.6 More on linear systems and Invertible Matrices ,1.7 Diagonal, Triangular and Symmetric matrices	1	1, 4
4	Determinants ,2.1 Determinants by Cofactor Expansion ,2.2 Evaluating Determinants by row reduction ,2.3 Properties of Determinants; Cramer's rule	2	1
5	Euclidean Vector Spaces ,3.1 Vectors in 2-space, 3-space and n-space, 3.2 Norm, Dot Product, and Distance in R_n , 3.3 Orthogonality of vectors, 3.4 The Geometry of Linear Systems, 3.5 Cross Product	3	2
6	General Vector Spaces, 4.1 Real Vector Spaces, 4.2 Subspaces	4	2
7	4.3 Linear Independence / Dependence,4.4 Coordinates and Basis for a vector space ,4.5 Change of Basis	4	2
8	4.6 Row space, Column Space and Null Space, 4.7 Rank and Nullity & Fundamental Matrix Spaces, 4.8 Rank and Nullity & Fundamental Matrix Spaces	4	2
9	Eigenvalues and Eigen vectors ,5.1 Eigenvalues and Eigenvectors, 5.2 Diagonalization of matrices	5	2
10	Inner Product Spaces, 6.1 Inner product spaces, 6.2 Angle and Orthogonality in Inner product Spaces	6	3
11	6.3 Gram-Schmidt Process, QR- Decomposition	6	3
12	General Linear Transformations ,1.8 Matrix Transformations ,4.9 Basic Matrix Transformations in R^2 & R^3 ,8.1 General Linear Transformations ,8.2 Compositions and Inverse Transformations,	8	4
13	8.3 Isomorphism ,8.4 Matrices for General Linear Transformations ,8.5 Similarity	8	4
14	7.1 Orthogonal Matrices,7.5 Hermitian, Unitary and Normal Matrices	7	1

***Reference book chapters are given in brackets**

Assessment Tools	Weightage
Quizzes	10.0%
Assignments	10.0%
Mid Term Exam	30.0%
Final Exam	50.0%